A Project Report
On

E-Commerce Customer Purchase Analysis

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PROBLEM STATEMENT

The objective of this project is to analyze customer purchase data from an e-commerce platform. The dataset consists information about various customers, products, quantities, prices, brands, and city locations.

The goal is to identify:
i) Customers who spend the most money
ii) Most popular products
iii) Cities generating the highest revenue

This analysis helps the marketing team make better decisions for promotions and customer targeting.

SOLUTION / APPROACH

The project was implemented using multiple tools to perform data analysis and visualization:

1. **SQL (MySQL Workbench):**

   - Created database and table

   - Imported CSV dataset (***Note: Initially the data was recorded in notepad and from there it was imported to goggle sheets and exported as .csv file)***

   - Performed queries to analyze customer spending, product popularity, and city revenue

2. **Microsoft Excel:**

   - Created Total Purchase column

   - Used Pivot Tables for product and city analysis

   - Generated bar charts and pie charts for visualization

3. **Python (Pandas & Matplotlib):**

   - Loaded dataset using Pandas

   - Performed grouping and aggregation

   - Created automated visualizations (bar chart and pie chart)

This multi-tool approach ensures accurate and comprehensive analysis.

IMPLEMENTATION

SQL QUERIES

```sql
CREATE DATABASE E_COMMERCE_DB;
USE E_COMMERCE_DB;

CREATE TABLE PURCHASES (
customer_id INT, customer_name VARCHAR(50), city VARCHAR(50), product VARCHAR(50), Brand VARCHAR(50), quantity INT, price INT);

SELECT * FROM purchases;

SELECT customer_name,
        SUM(quantity * price) AS total_spent
FROM purchases
GROUP BY customer_name
ORDER BY total_spent DESC;

SELECT product,
        SUM(quantity) AS total_quantity
```

```sql
FROM purchases
GROUP BY product
ORDER BY total_quantity DESC;

SELECT city,
        SUM(quantity * price) AS revenue
FROM purchases
GROUP BY city
ORDER BY revenue DESC;

SELECT brand,
        SUM(quantity * price) AS brand_revenue
FROM purchases
GROUP BY brand
ORDER BY brand_revenue DESC;
SELECT COUNT(*) FROM purchases;
```

i) Which customers spend the most money on the platform?

This query calculates the total amount spent by each customer by multiplying quantity and price for each purchase and aggregating the results. The output is sorted in descending order to identify the highest spending customers.

From the analysis, AMAN is the highest spending customer, followed by Rahul and Karan. These customers can be considered high-value customers and can be targeted for premium services and promotional offers.

customer_name	total_spent
AMAN	400000
RAHUL	310000
KARAN	199000
KRISHNA	165000
PRIYA	160000
HARSH	160000
PARTH	130000
ARJUN	103000
HARRY	100000
AKASH	99900
MEENA	90000
NEHA	80000
RAJ	80000
YASH	64000
RITU	60000
AMIT	60000
SNEHA	51000
AMAR	38000

Total Spending by Customers

ii) Which brands generate the highest revenue on the platform?

This query calculates the total revenue generated by each brand by aggregating the product of quantity and price. The results are grouped by brand and sorted to identify the highest revenue-generating brands.

From the analysis, Apple generates the highest revenue, followed by Dell and Samsung. Brands like Boat and Realme contribute comparatively lower revenue. This insight helps identify strong-performing brands and supports strategic decisions such as promotions and inventory planning.

Result Grid | Filter Rows:

brand	brand_revenue
APPLE	796900
DELL	460000
SAMSUNG	307000
VIVO	165000
ONEPLUS	160000
HP	141000
LENOVO	130000
LENOVO	100000
REALME	80000
BOAT	10000

Brand-wise Revenue Analysis

Result Grid | Filter Rows:

city	revenue
BANGALORE	604900
KOLKATA	364000
CHANDIGARH	300000
MUMBAI	291000
DELHI	288000
HYDERABAD	224000
PUNE	158000
CHENNAI	60000
BHUBANESWAR	60000

City-wise Revenue Analysis

iii) Which cities generate the highest revenue on the platform?

This query calculates the total revenue generated from each city by summing the product of quantity and price for all transactions. The results are grouped by city and sorted in descending order to identify the highest revenue-generating locations.

From the analysis, Bangalore generates the highest revenue, followed by Kolkata and Chandigarh. Cities like Chennai and Bhubaneswar contribute the least revenue. This indicates that metropolitan cities contribute significantly more to overall sales.

iv) Which products are most popular based on quantity sold?

This analysis identifies the most popular products based on quantity sold, helping businesses focus on high-demand items.

Result Grid | Filter Rows:

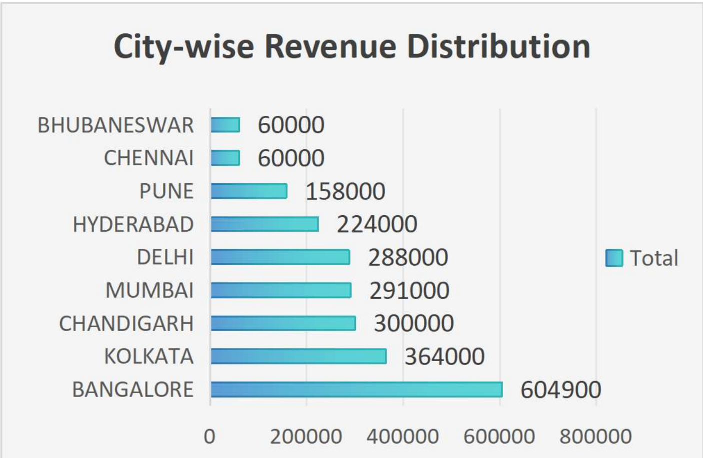
product	total_quantity
MOBILE	8
SMARTWATCH	7
LAPTOP	6
TABLET	6
FITNESS RING	4
I-PHONE	2
HEADPHONES	2
AIRPODS	2
i-PAD	1
MACBOOK	1

Pivot tables and Chart representation

product	Sum of quantity
AIRPODS	2
FITNESS RING	4
HEADPHONES	2
i-PAD	1
I-PHONE	2
LAPTOP	6
MACBOOK	1
MOBILE	8
SMARTWATCH	7
TABLET	6
Grand Total	**39**

Pivot table for Product Sales

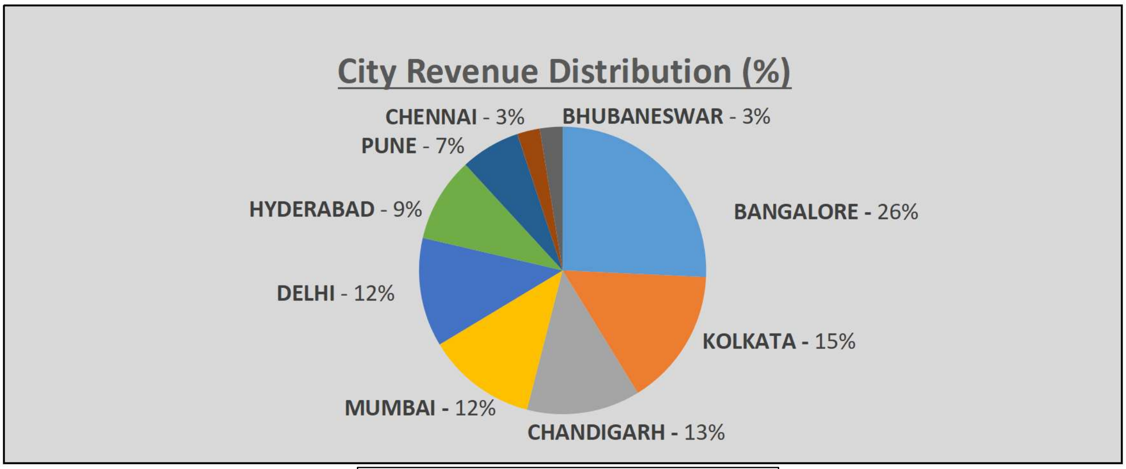
Bar Chart Representation



Bar Chart Representation

city	Sum of Total_Purchase
BANGALORE	604900
KOLKATA	364000
CHANDIGARH	300000
MUMBAI	291000
DELHI	288000
HYDERABAD	224000
PUNE	158000
CHENNAI	60000
BHUBANESWAR	60000
Grand Total	**2349900**

Pivot table for City-Wise Revenue Distribution



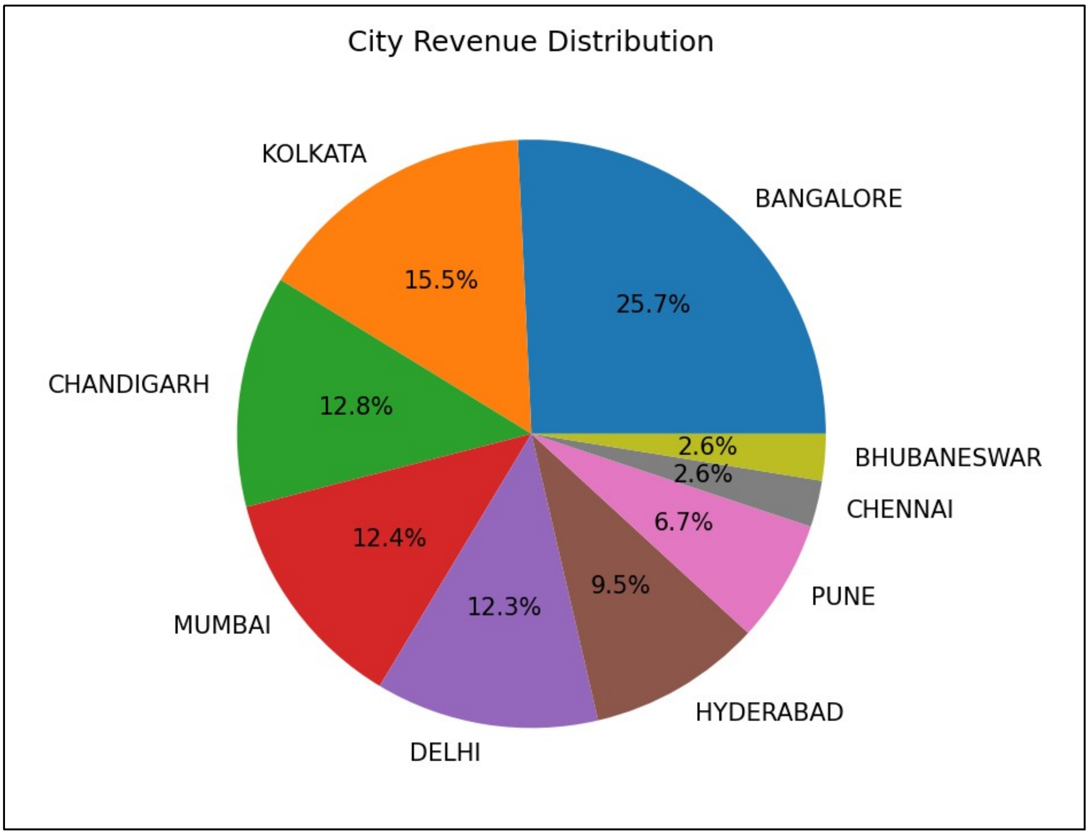
Pie chart Representation

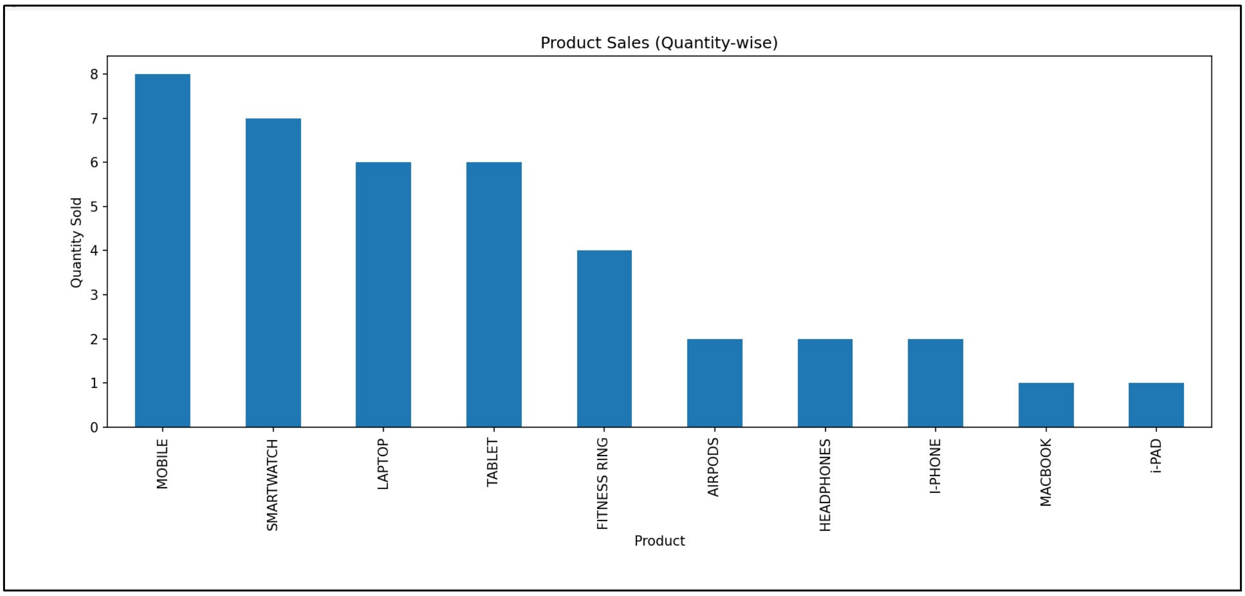
Python Output

```
KIIT0001@BT05042A MINGW64 ~/Desktop/data fundamentals/PROJECT/E_Commerce
$ C:/Users/KIIT0001/AppData/Local/Programs/Python/Python312/python.exe "
Top Customers:

customer_name
AMAN       400000
RAHUL      310000
KARAN      199000
KRISHNA    165000
PRIYA      160000
HARSH      160000
PARTH      130000
ARJUN      103000
HARRY      100000
AKASH       99900
MEENA       90000
NEHA        80000
RAJ         80000
YASH        64000
RITU        60000
AMIT        60000
SNEHA       51000
AMAR        38000
Name: total_purchase, dtype: int64
```

```
Product Popularity:

product
MOBILE          8
SMARTWATCH      7
LAPTOP          6
TABLET          6
FITNESS RING    4
 AIRPODS        2
HEADPHONES      2
I-PHONE         2
MACBOOK         1
i-PAD           1
Name: quantity, dtype: int64
```

```
City Revenue:

city
BANGALORE      604900
KOLKATA        364000
CHANDIGARH     300000
MUMBAI         291000
DELHI          288000
HYDERABAD      224000
PUNE           158000
CHENNAI         60000
BHUBANESWAR     60000
Name: total_purchase, dtype: int64
```


City Revenue Distribution
KOLKATA
BANGALORE
CHANDIGARH
BHUBANESWAR
CHENNAI
PUNE
HYDERABAD
DELHI
MUMBAI
15.5%
25.7%
12.8%
2.6%
2.6%
6.7%
9.5%
12.3%
12.4%


Product Sales (Quantity-wise)
Quantity Sold
0
1
2
3
4
5
6
7
8
MOBILE
SMARTWATCH
LAPTOP
TABLET
FITNESS RING
AIRPODS
HEADPHONES
I-PHONE
MACBOOK
i-PAD
Product

KEY INSIGHTS

- Mobile is the most popular product with the highest quantity sold.
- Bangalore generates the highest revenue (26% contribution).
- Kolkata and Chandigarh also contribute significantly to total revenue.
- Chennai and Bhubaneswar show the lowest revenue contribution.
- Product demand is higher for smart and portable electronic devices.

These insights can help businesses focus on high-performing products and cities while improving strategies for low-performing regions.

TECH STACK

- MySQL Workbench (Database & SQL Analysis)
- Microsoft Excel (Data Analysis & Visualization)
- Python (Pandas, Matplotlib)
- Visual Studio Code (Development Environment)

UNIQUE FEATURES

- Integration of SQL, Excel, and Python in a single project
- Automated data analysis using Python
- Visual representation of insights using charts
- Inclusion of brand-based analysis for deeper insights

FUTURE IMPROVEMENTS

- Integration with real-time data sources
- Development of interactive dashboards using Power BI or Tableau
- Customer segmentation using machine learning
- Predictive analysis for future sales trends
- Automated data validation techniques can also be incorporated to ensure consistency during data entry.***(Eg. Iphone and Smartphones can be categorized further into Mobile Group)***